Depreciation

1 Ray purchased a $\$9300$ computer, which depreciates at a compounded rate of 25% p.a. Answer the following questions without using the depreciation formula.

- **a** What is the salvage value of the computer after the first year?
- **b** By how much did the computer depreciate in the first year?
- **c** What is the salvage value after the second year?
- **d** By how much does computer depreciate over the second year only?
- **e** What is the total depreciation over the 2 years?

2 Neil purchased a $\$1300$ motorbike which depreciates at a compounded rate of 10% p.a.

- **a** Find the total depreciation over the two years.
- **b** Find the percentage of the original value remaining after two years. Round your answer to the nearest percentage.

3 A car was originally purchased for $\$3000$, and depreciates at 14% p.a. Use the depreciation formula to calculate its expected value after 8 years.

4 A dishwasher selling for $\$800$, depreciates at 11% p.a.

- **a** What percentage of the original value will remain after 1 year?
- **b** What percentage of the original value will remain after 2 years? Round your answer to two decimal places.
- **c** What percentage of the original value will remain after 3 years? Round your answer to two decimal places.
- **d** How many full years will it take for the dishwasher to lose half its original value?
- **e** How many years will it take for the dishwasher to lose 90% of its original value?

5 After one year, the value of a company's machinery had decreased by $\$3900$ from $\$65\,000$.

- **a** At what rate did the machinery depreciate in the first year?
- **b** If the value of the machinery continues to decrease at this rate every year, what will it be worth in another 19 years?

6 A car depreciated in value from $\$20\,000$ to $\$17\,530$ in 3 years. Find the annual depreciation rate, as a percentage to two decimal places.

7 Han's share portfolio of $\$83\,000$ decreased in value by 14% per month for the first 4 months of the Global Financial Crisis and then 3% per month for the 5 months after that. Find the value of his portfolio after 9 months.

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8 A laptop depreciated by 21% p.a. and was valued at $\$700$ after 9 years. Find its original price.

9 A microwave that costs $\$700$ depreciates at 10% p.a. After how many full years will the value of the microwave be less than:

a $\$525$ **b** $\$350$ **c** $\$70$

10 Joanne purchased a laptop for $\$800$, which depreciates at 13% p.a. Find its salvage value after 6 years.

11 A camera originally purchased for $\$3000$, depreciates at a rate of 29% p.a. Use the declining-balance depreciation formula to calculate its expected value after 10 years.

12 Fred deposited $\$9000$ into a stock portfolio. This amount decreased by 11% each year for 10 consecutive years. What was the value of the portfolio after 10 years?

13 The government wants to decrease its spending on job creation. Currently, it is spending $\$11$ million and will decrease it by 2.55% p.a. over the next 8 years. Calculate the government's spending in 8 years' time. Round your answer to the nearest dollar.

14 The population of some native bees is declining at a rate of 8% per year. If there are $10\,000$ bees in a hive now, how many will there be in 5 years?

15 Eileen purchased a new bed for $\$8000$, which depreciates at the rate of 15% p.a. Answer the following questions by repeated multiplication.

a What is the expected value of the bed after 4 years?

b What is the total depreciation of the bed after 4 years?

c What is the percentage of the original value that will remain after 4 years? Give your answer as a percentage to two decimal places.

16 Marge purchased new office equipment for a total of $\$6560$.

a Calculate the value of the equipment after 3 years if she uses straight line depreciation at 7% per year.

b Calculate the value of the equipment after 3 years if she uses declining-balance depreciation at 7% per year.

c Assuming declining-balance depreciation, after how many full years will the equipment be worth $\$3000$?

17 A Macbook currently worth $\$1823$ was purchased 3 years ago for $\$2000$.
Find the annual depreciation rate, R. Write your answer as a percentage to two decimal places.

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18 The population of an island was $1\,610\,000$ in 1950. Due to rising sea levels leaving more and more parts of the island underwater, the population fell at the rate of 18% p.a. What was the population in 1989? Write your answer to the nearest whole number.

19 A radioactive substance decays by 1.86% each year and there are currently $19\,300$ grams of the substance present. If there were P grams present 10 years ago, find the value of P to one decimal place.

20 Mae purchased an iPod for $\$100$, which depreciates at 18% p.a. What is its resale value after 4 years?

21 A car depreciated in value from $\$20\,000$ to $\$17\,530$ in 3 years.
At what rate per annum, R, did it depreciate? Write your answer as a percentage correct to two decimal places.

22 A swimming pool is losing water at the rate of 8% per week due to evaporation. The pool currently holds 490 kilolitres of water.

- **a** How much water will be lost in the next 26 days? Give your answer in kilolitres and correct to two decimal places.
- **b** How many weeks will it take for the pool to become only half full?

23 Viewer ratings for a show increased by 5% per episode for the first 7 episodes, and then decreased by 6% per episode for the last 6 episodes. If there were $129\,000$ viewers during the first episode, how many viewers were there during the last episode? Round your answer to the nearest viewer.

24 The number of people with access to basic healthcare is decreasing by 2.4% of the previous year's number. If $1\,414\,000$ people currently have access, how many people will have access to basic healthcare in 7 years? Round your answer to the nearest person.

25 A lab has $\$1600$ worth of radioactive polonium stored away, which depreciates at 0.81% p.a. as it decays. What value of polonium does the lab have after 86 years?

26 After 1 year, the value of a company's machinery had decreased by $\$8946$ from $\$63\,900$.

- **a** At what rate did the machinery depreciate in the first year? Give your answer as a percentage.
- **b** If the value of the machinery continues to decrease by the same rate every year, what will it be worth in another 11 years?

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27 Amelia purchased a signed photograph for $\$756$, which depreciates at 4.34% p.a. After 8 years, using single line depreciation, the signed photograph would be worth $\$493.52$.

a Calculate its resale value after 8 years using the declining-balance method.

b Which method of depreciation should Amelia hope her signed photograph follows: straight line method or declining-balance method?

28 Sean is looking to purchase a car that he is planning to re-sell in 3 years. A new luxury car is going to cost $\$10\,000$, which depreciates at 20% p.a. A reliable used car costs $\$5000$ and depreciates at 10% p.a.

a Calculate the resale value after 3 years of the:

i Luxury car. **ii** Used car.

b Calculate the loss in value after 3 years of the:

i Luxury car. **ii** Used car.

c Which car has the higher resale value?

d Which car has the smallest decrease in value?

e Which car should Sean purchase to minimise his losses?

29 A computer depreciated by 16% p.a. and was valued at $\$517$ after 6 years. The owner of the computer wants to calculate the original cost.

a To find the initial value, V_0, the following equation needs to be rearranged:

$$S = V_0\,(1 - r)^n$$

Write the formula for V_0.

b What was the computer's original price?

30 A property depreciated by 4.1% p.a. and was valued at $\$901\,500$ after 29 years. What was its original price?

31 James is purchasing scuba diving gear that costs $\$600$ and depreciates at 14% p.a. After how many full years will the value of the scuba diving gear be less than:

a $\$450$? **b** $\$300$? **c** $\$60$?

32 A ball is dropped from a height of $21\ \text{m}$. After each bounce, it reaches half the previous height. Find the height reached after the 5th bounce correct to two decimal places.

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33 A car salesman received a total commission of $\$3496$ at the beginning of the month. He expects that since fewer cars are being purchased, his commission will decrease by 12% each month over the next few months. According to his prediction, what will be his commission in 7 months' time?

34 Cinema ticket sales for a new movie increased by 3% per day for the first 5 days after its release date, and then decreased by 4% per day for the next 3 days. If $\$259\,000$ in ticket sales were made on the release date, how much was made in sales on the 8th day after its release to the nearest dollar?

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